

Lesson 15

Topic: Movies Analyses Dashboard with DAX

Prerequisites: All data is provided in **Movies.xlsx**

1. Load **Movies.xls** into Power BI desktop

Step-by-Step: Load **Movies.xlsx** into Power BI Desktop

Step 1: Open Power BI Desktop

- Launch **Power BI Desktop** on your computer.

Step 2: Load the Excel File

1. Click **Home** tab → Click “**Excel Workbook**” (under “Data” section).
2. Browse and select the file **Movies.xlsx**.
3. Click **Open**.

Step 3: Navigator Pane

- The Navigator window will show all the **sheets or tables** in the Excel file.
- Select all necessary sheets or tables (e.g., Movies, Ratings, Genres, etc. depending on what’s inside).
- Click **Load** (or **Transform Data** if you want to clean it first).

Optional: Data Cleanup (in Power Query)

If needed:

- Remove blank columns/rows.
- Change data types (e.g., Date columns, Number columns).
- Rename column headers for clarity.

Once done, click **Close & Apply** to load into the Power BI model.

Once the data is loaded, you're ready to begin:

Creating relationships

Writing DAX measures (like Avg Rating, Total Box Office, etc.)

Designing your **Movie Analytics Dashboard**

2. **Remove or impute missing values in Budget and Box Office (e.g., replace with 0 or median)**

Step-by-Step: Handle Missing Values in Budget and Box Office

Step 1: Open Power Query

1. In Power BI, go to the **Home** tab.
2. Click on “**Transform Data**” → This opens **Power Query Editor**.

Step 2: Select Your Movie Table

- On the left pane, select the table that contains the Budget and Box Office columns (e.g., Movies)

Step 3: Check for Missing Values

- In the Budget and Box Office columns:
 - Missing values usually appear as null.

Option A: Replace Missing Values with 0

1. Right-click the Budget column → Select **Replace Values**.
2. In the dialog:
 - **Value to Find:** leave empty
 - **Replace With:** 0
3. Repeat for the Box Office column.

Option B: Replace Missing Values with Median

1. Right-click the Budget column → Select “**Add Column**” → then click “**Statistics**” → “**Median**”.
2. Note the median value it returns.
3. Use the **Transform** tab → **Replace Values** to replace null with that median value manually.
4. Repeat for Box Office.

*Power Query does not have a built-in median imputation by default, so you must calculate it manually or do it using **DAX** after loading.*

Step 4: Apply Changes

- Click “**Close & Apply**” in the Power Query Editor to load the cleaned data into Power BI.

3. Ensure all columns are in a appropriate format.

Step-by-Step: Set Appropriate Column Formats in Power Query

Step 1: Open Power Query Editor

- In Power BI → Go to **Home tab** → Click “**Transform Data**”.

Common Data Type Assignments for Movie Datasets

Column Name	Suggested Data Type
Movie Title	Text
Genre	Text
Director	Text
Release Date	Date
Year	Whole Number
Duration (minutes)	Whole Number
Budget	Decimal Number (or Whole if always integers)
Box Office	Decimal Number
Rating (e.g., IMDb)	Decimal Number
Votes	Whole Number
Country	Text

Step 2: Change Data Types

For each column:

1. Click the **data type icon** next to the column name (like ABC123, Calendar, etc.).
2. Select the appropriate type from the dropdown.

3. Or right-click column → choose **Change Type**.

Step 3: Check for Errors

- After changing types, check if any values turn into “**Error**”.
- If yes, go to “**View**” → **Enable Column Quality** to help locate and fix issues.

Step 4: Apply the Changes

- Once all data types are correct → Click “**Close & Apply**”.

Tip:

If Budget or Box Office were originally text (like "\$200M" or "USD 1500000000"), you'll need to:

- **Remove text/symbols** first (e.g., using **Replace Values** or **Split Column by Delimiter**).
- Then convert the cleaned result to **Decimal Number**.

4. **Create a separate DATE table by using Release Date column, use Addcolumns, Calendarauto, filter functions and make sure a relationship exist in the data model view between two tables.**

Step-by-Step: Create a Date Table from Release Date

Step 1: Create Date Table (Using DAX)

In Power BI:

1. Go to **Modeling** tab
2. Click “**New Table**”
3. Paste this DAX:

DAX

CopyEdit

DateTable =

```
ADDCOLUMNS(  
    CALENDARAUTO(),  
    "Year", YEAR([Date]),
```

```

    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Year-Month", FORMAT([Date], "YYYY-MM")
)

```

CALENDARAUTO() will auto-detect the min/max Release Date
 ADDCOLUMNS adds useful columns like Year, Month, Quarter

Optional: Add Filter (to restrict future dates)

If your data has weird future dates, filter them like this:

DAX

CopyEdit

DateTable =

VAR BaseCalendar = CALENDARAUTO()

RETURN

ADDCOLUMNS(

 FILTER(BaseCalendar, [Date] <= TODAY()),

 "Year", YEAR([Date]),

 "Month Number", MONTH([Date]),

 "Month Name", FORMAT([Date], "MMMM"),

 "Quarter", "Q" & FORMAT([Date], "Q"),

 "Year-Month", FORMAT([Date], "YYYY-MM")

)

Step 2: Create a Relationship in Model View

1. Go to **Model View**
2. Drag Date from DateTable → onto Release Date in Movies table.
3. Ensure the relationship is:
 - One-to-Many

- Single direction (from DateTable → Movies)

Now Power BI will correctly understand the time logic.

Why this is important:

- Using a **separate Date table** is best practice (vs. relying on Auto Date/Time).
- Enables advanced time intelligence like:
 - YTD, QTD, YoY comparisons
 - Custom calendar filters (e.g., only show Decembers)
 - Better performance with DAX

5. Create the following DAX measures and calculated columns to practice filter contexts, aggregations, and time intelligence.

- **Create a calculated column called profit by using Budget and Box office column**
- **Create a calculated column to categorize movies by run time for segmentation. (Run Time < 90, Short Run Time > 90 and Run Time < 120, Medium Run Time > 120 , Long) Create following measures:**
 - **Total Box Office**
 - **Average Budget**
 - **Average Margin**
 - **Total movies with Oscars**
 - **Top Genre by Box Office**
 - **Year-over-Year Box Office Growth**
 - **Average Nominations per Director**

PART 1: CALCULATED COLUMNS

1. Profit

DAX

Profit = Movies[Box Office] - Movies[Budget]

2. Runtime Category

DAX

Runtime Category =

SWITCH(

TRUE(),

Movies[Run Time] < 90, "Short",

Movies[Run Time] >= 90 && Movies[Run Time] < 120, "Medium",

Movies[Run Time] >= 120, "Long",

BLANK()

)

PART 2: MEASURES

3. Total Box Office

DAX

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Total Box Office = SUM(Movies[Box Office])

4. Average Budget

DAX

Average Budget = AVERAGE(Movies[Budget])

5. Average Margin (Box Office - Budget)

DAX

Average Margin = AVERAGE(Movies[Box Office] - Movies[Budget])

6. Total Movies with Oscars

Assuming you have a column like Oscar Wins:

DAX

Total Movies with Oscars =

CALCULATE(

```
COUNTROWS(Movies),  
FILTER(Movies, Movies[Oscar Wins] > 0)  
)
```

7. Top Genre by Box Office

This returns the top-selling genre (text value):

DAX

Top Genre by Box Office =

VAR GenreTable =

```
SUMMARIZE(  
    Movies,  
    Movies[Genre],  
    "TotalRevenue", SUM(Movies[Box Office])  
)
```

VAR MaxRevenue = MAXX(GenreTable, [TotalRevenue])

RETURN

```
CALCULATE(  
    MAX(Movies[Genre]),  
    FILTER(GenreTable, [TotalRevenue] = MaxRevenue)  
)
```

8. Year-over-Year Box Office Growth

Make sure your DateTable is related to Movies[Release Date].

DAX

YoY Box Office Growth =

```
VAR CurrentYearRevenue = CALCULATE(SUM(Movies[Box Office]),  
USERRELATIONSHIP(DateTable[Date], Movies[Release Date]))
```

```
VAR LastYearRevenue = CALCULATE(SUM(Movies[Box Office]),
USERRELATIONSHIP(DateTable[Date], Movies[Release Date]),
SAMEPERIODLASTYEAR(DateTable[Date]))

RETURN

    DIVIDE(CurrentYearRevenue - LastYearRevenue, LastYearRevenue)
```

9. Average Nominations per Director

Assuming a Nominations column:

DAX

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Average Nominations per Director =

```
AVERAGEX(
    SUMMARIZE(Movies, Movies[Director], "TotalNoms",
SUM(Movies[Nominations])),
    [TotalNoms]
)
```

Summary

Task	DAX Type	Name
Profit Calculation	Calculated Column	Profit
Runtime Segmentation	Calculated Column	Runtime Category
Box Office Total	Measure	Total Box Office
Budget Average	Measure	Average Budget
Margin Average	Measure	Average Margin
Count Oscars	Measure	Total Movies with Oscars
Highest Box Office Genre	Measure	Top Genre by Box Office
Year-over-Year Growth	Measure	YoY Box Office Growth

Avg. Nominations by Director Measure Average Nominations per Director

6. Visualization Requirements: Create an interactive dashboard with the following visuals to practice Power BI's visualization features.

- **Page 1: Overview Dashboard. Recommended Visuals:**
 - **Card: Total Box Office, Profit Margin, Movies with Oscars.**
 - **Bar Chart: Total Box Office by Genre (stacked by Certificate).**
 - **Line Chart: Box Office trend by Release Year.**
 - **Slicer: Filter by Country and Release Date (range).**
 - **KPI: YoY Box Office Growth (target: >0%).**
- **Page 2: Director Analysis. Recommended Visuals:**
 - **Treemap: Budget by Director (size by Budget, color by Oscar Wins).**
 - **Table: List Directors, Total Nominations, Total Oscars, Avg Nominations per Director.**
 - **Slicer: Filter by Genre.**
 - **Donut Chart: Run Time Category distribution for selected Director.**
- **Page 3: Genre and Country Insights Recommended Visuals:**
 - **Matrix: Genre vs. Country with Total Box Office as values.**
 - **Pie Chart: Share of Box Office by Certificate.**
 - **Custom Visual: Word Cloud for Genre (requires marketplace visual).**
 - **Slicer: Filter by Run Time Category.**
 - **Use conditional formatting in the matrix (e.g., color scale for Box Office).**

PAGE 1: Overview Dashboard

Goal: Executive Summary

Visuals to Include

1. Cards (Top Row)

- Total Box Office
- Average Margin or Profit Margin
- Total Movies with Oscars

Use **large, clear Card visuals** with color-coded data labels

Optional: Use Data Bars inside cards using custom visuals

2. Bar Chart – Total Box Office by Genre, **stacked by Certificate**

- Axis: Genre
- Value: Box Office
- Legend: Certificate (e.g., PG-13, R)

3. Line Chart – Box Office Trend by Release Year

- Axis: DateTable[Year]
- Value: Total Box Office
- Optional: Add tooltip for YoY Growth

4. Slicers (Sidebar or Top)

- Country (Dropdown or vertical list)
- Release Date (Between slider)

5. KPI Visual – YoY Box Office Growth

- Use KPI visual from native visuals
- Target = 0%
- Indicator: Red (negative), Green (positive)

PAGE 2: Director Analysis

Goal: Deep Dive into Directors

Visuals to Include

1. Treemap – Budget by Director

- **Size** = Budget
- **Color** = Oscar Wins (gradient or diverging scale)

2. **Table (Summary)**

Columns:

- Director
- Total Nominations
- Total Oscars
- Avg Nominations per Director

Use DAX measures for these. Apply **sorting and formatting**.

3. **Slicer** – Filter by Genre

4. **Donut Chart** – Runtime Category Distribution

- Values: Count of Movies
- Legend: Runtime Category
- Filter interaction: Applies to selected director

PAGE 3: Genre & Country Insights

Goal: Explore Patterns by Region & Genre

Visuals to Include

1. **Matrix** – Genre vs. Country

- Rows: Genre
- Columns: Country
- Values: Total Box Office
- Use **Conditional Formatting (Color Scale)** on values

2. **Pie Chart** – Share of Box Office by Certificate

- Legend: Certificate
- Value: Total Box Office

3. **Custom Visual: Word Cloud (Genre)**

- Import from marketplace: Word Cloud by Microsoft
- Field: Genre

4. **Slicer** – Runtime Category

Best Practices & UX Tips

- Enable **Page Navigation** buttons or use Tabs (Bookmarks).
- Group visuals using **shapes or grid containers**.
- Enable **tooltips** to show detailed info (like Margin, Votes, Ratings).
- Use **Themes** to maintain consistent color and font.
- Turn off unwanted interactions where needed (Format → Edit Interactions).